

INF2007 Mobile App Development

Rehab Mate

ZAPJOLLY

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Introduction

Rehabilitation

- Helps individuals recover from injuries through structured exercises
- Monitoring progress and receiving feedback is crucial for successful rehab

Problem Statement

- Many patients struggle with rehab due to:
 - Busy schedules
 - High session costs
 - Mobility limitations
- Delayed recovery and improper healing



RehabMate

- **Personalized** and **accessible** rehabilitation
- Uses **motion sensors** to track exercises
- **Supports wide range** of rehab exercises
- User management, Progress tracking, Appointment

Why InHouse Rehab?

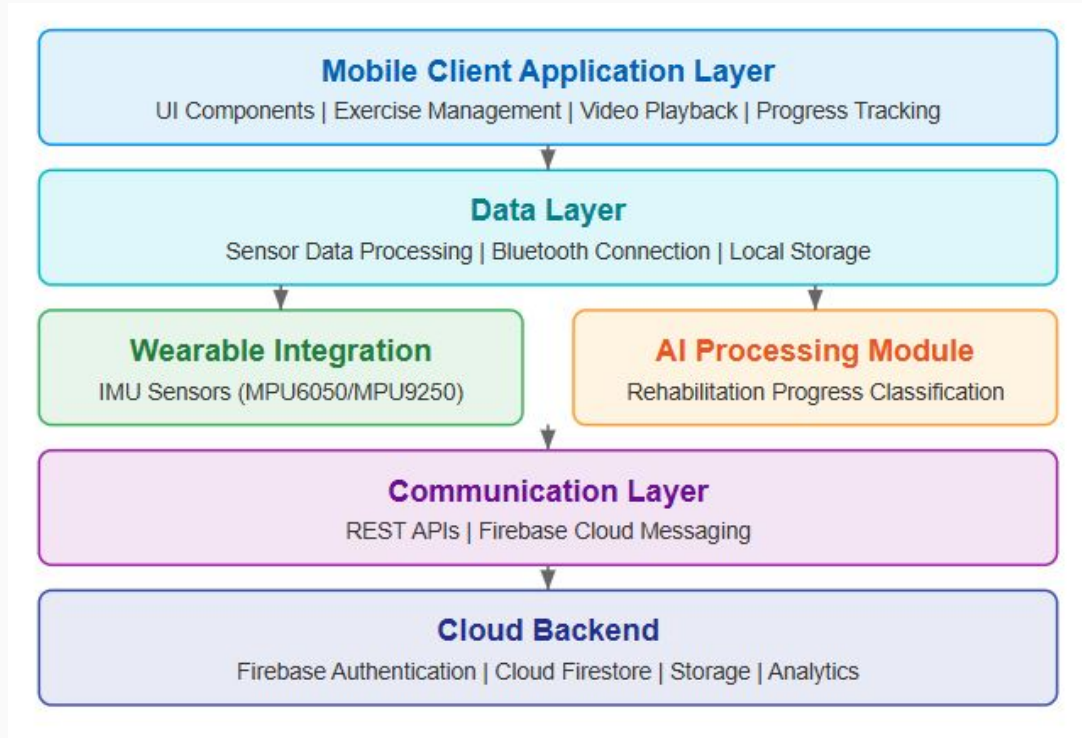
- **Reduces reliance** on frequent clinic visits
- **Reduces** risk of **re-injury**
- **Reduces** physiotherapy **costs**



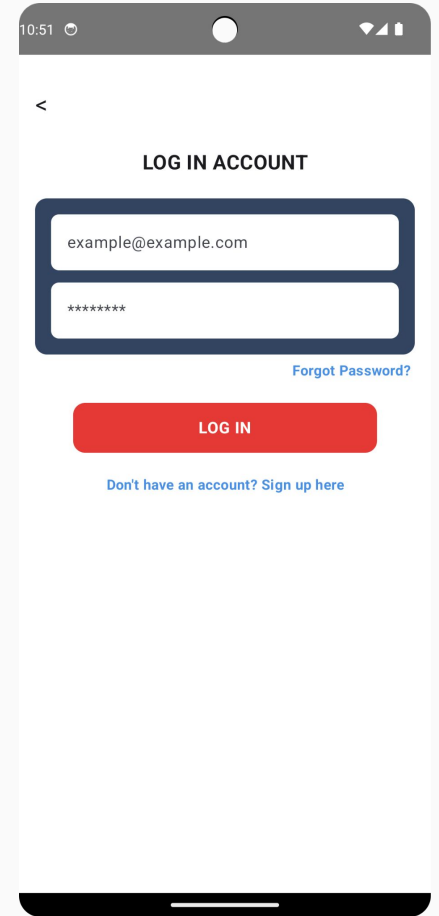
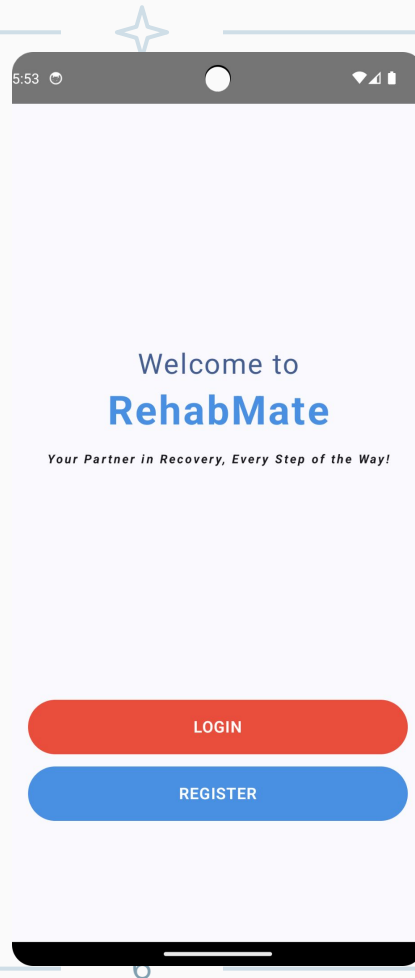
Key Aspects

1. **Mobile AI:** Predicts user's progress based on their age, number of activities, total activities duration and their overall progress
2. **Mobile Multimedia:** Uses audio for hands-free exercise guidance
3. **Mobile Sensors:** Collects user's rotation, gyroscope & heartbeats during exercise, for logging and ensuring patient doing it correctly
4. **Mobile Communication:** Enables real-time transmission of sensor data from wearable watch to phone

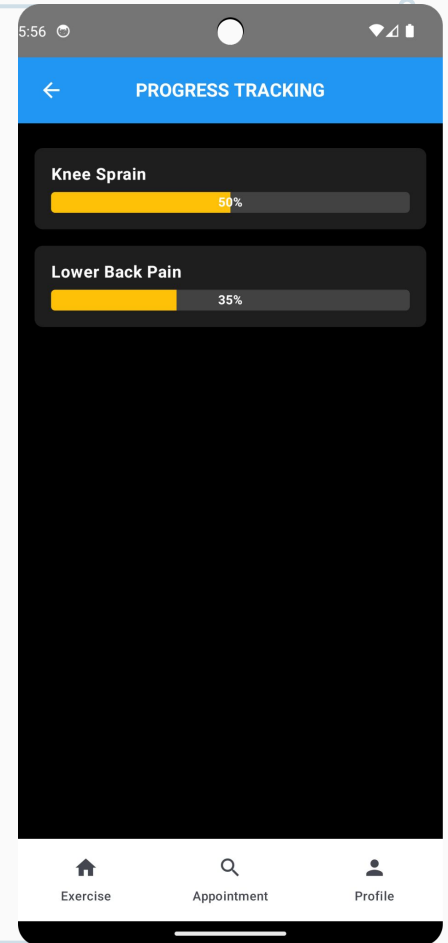
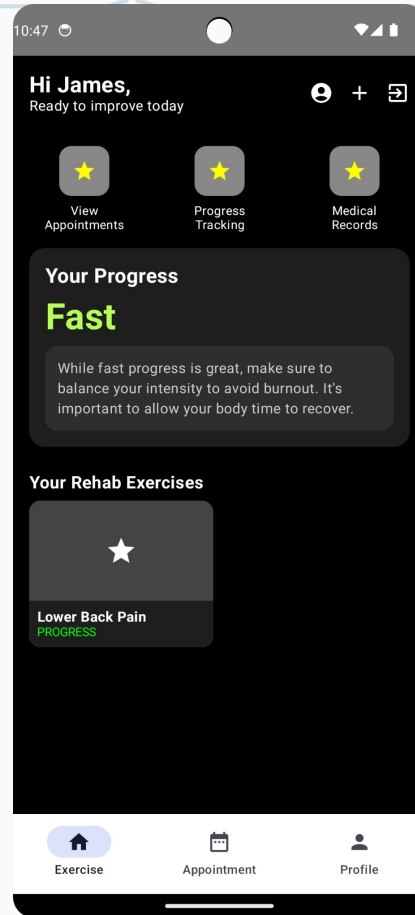
System Architecture



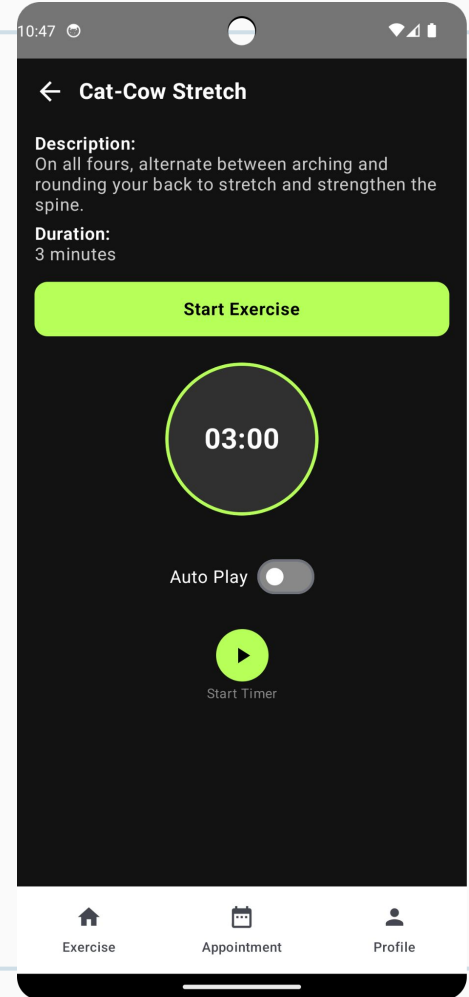
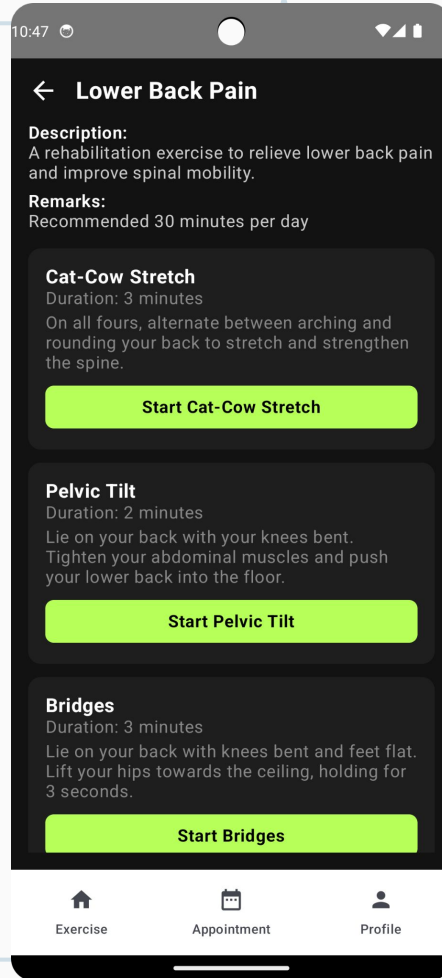
Prototype



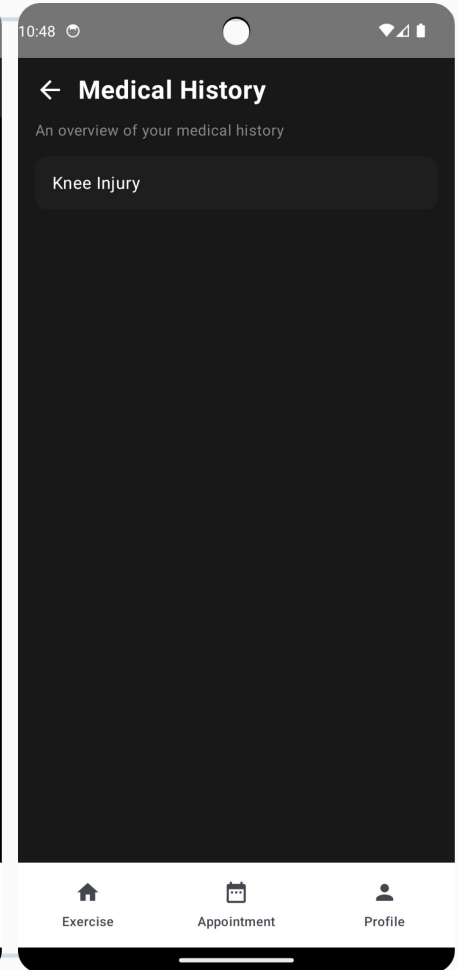
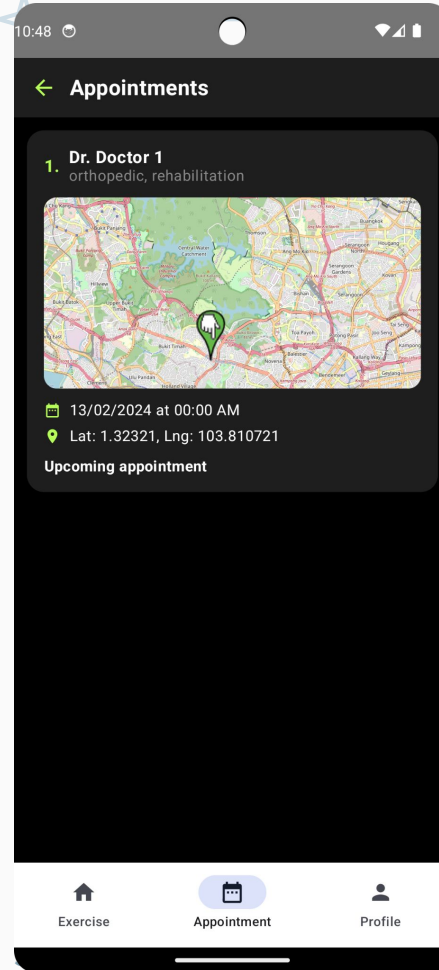
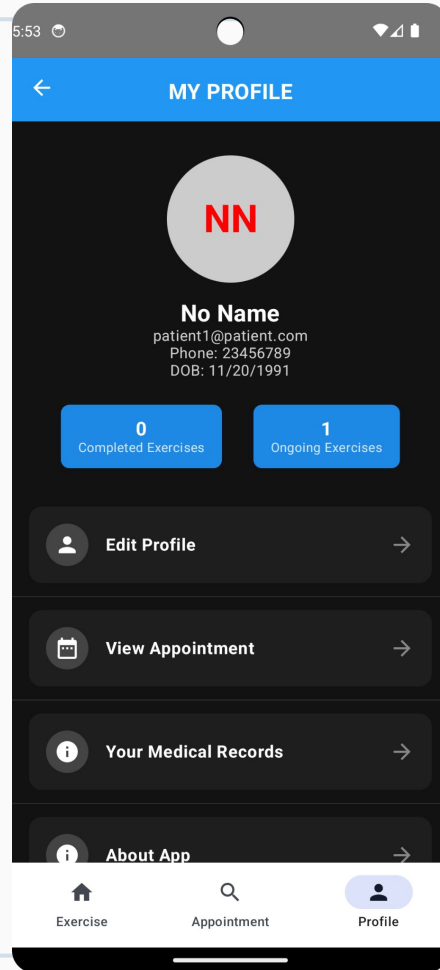
Prototype



Prototype



Prototype

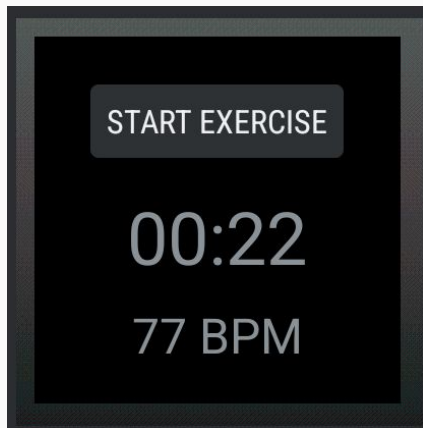




```
Wear OS Square (emulator-5556) Android 14, API 34 package:mine
152 WearOS com.example.rehabmate D Sent sensor data to mobile: 817f6b64
152 RotationData com.example.rehabmate D Rotation (Real) -> X: 0.67761487, Y: -4.1598472E-5, Z: 2.6331353E-4
190 EGL_emulation com.example.rehabmate D app_time_stats: avg=999.75ms min=999.75ms max=999.75ms count=1
152 GyroData com.example.rehabmate D Gyro (Simulated) -> X: -1.77, Y: -1.91, Z: -0.06
152 HeartRateData com.example.rehabmate D Heart Rate (Simulated): 167 BPM
190 EGL_emulation com.example.rehabmate D app_time_stats: avg=500.72ms min=101.14ms max=900.29ms count=2
152 WearOS com.example.rehabmate D Sent sensor data to mobile: 817f6b64
```

```
Pixel 8 (emulator-5554) Android 14, API 34 package:mine
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xxxxxx statsUid=-1
xxxxxx statsUid=-1
GC freed 40478(1934KB) AllocSpace objects, 1(24KB) LOS objects, 49% free, 8042KB/15MB, paused 1.919ms,136us total 203.600ms
-> {"gyro":{"x":-4.5902023,"y":-4.84984,"z":-2.4221663},"rotation":{"x":0.6775667,"y":-3.1822703E-5,"z":2.587346E-4},"heartRate":69}
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```

Prototype





Conclusion

- **Seamless integration** on mobile AI, communication, sensors, and multimedia
- **Increased Accessibility:** Reduces clinic visits → Convenient rehab & reduced costs

Future Improvements

1. Integration of rehab videos and AI-based pose recognition and insights
2. Cloud integration



Individual Contributions

Amanda

1. Implemented Firebase APIs to fetch data and manage authentication.
2. Designed Firestore schema
3. UI page's
4. Integration of API's
5. Reports and slides

Jana

1. **AI:** Machine learning (Training & Deployment to application)
2. **UI:** Medical History and AI progress
3. **Backend:** Firebase data retrieval
4. Slides and report

Putri

1. **Sensor:** Wear OS application, retrieval and send IMU Data to mobile
2. **UI :** Wear OS & Progression Tracking
3. **Backend:** Firebase data retrieval
4. Slides and report



Individual Contributions

Sean

1. **AI:** Assist on researching how to implement
2. **UI:** Profile page, Exercise screen, visual bugs
3. **Backend:** Fetching data, API
4. Slides and report

Zuhairi

1. **Git:** Developed baseline code
2. **UI:** Refined authentication screens
3. **Backend:** Firebase create and read functionalities
4. Slides and report



Thank you!